Lecture 9: Game Theory

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Lecture overview

- Simultaneous single-shot (single-move) games
- Repeated simultaneous games
- Sequential games

Oligopoly

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- Firms in an oligopoly depend on each other's decisions.
- Each firm must think about how rivals may react before choosing price, output, or advertising.
- A firm's profit depends not only on its own decision, but also on competitors' decisions.
- Firms often make assumptions about rivals' likely behaviour.

Game Theory

- Game theory helps explain strategic decision-making.
- It shows how firms choose the best strategy when their actions affect each other.
- How economists use game theory in oligopoly
    - To study how firms set prices
    - To study how firms choose output/quantity
    - To study how firms decide on advertising levels

Game Theory

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- Game theory beyond economics
  - Used by economists, political scientists, and military analysts.
  - Useful for analysing any situation involving strategic interaction.
- Real-world applications
  - Negotiations over Greek debt in 2015
  - Negotiations over the UK’s exit from the European Union
  - Bargaining between unions and management
  - Bargaining between a buyer and seller, such as in car sales
  - Conflicts between polluters and people harmed by pollution

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What exactly is 'game theory'?

18 February 2015

The Greek Finance Minister Yanis Varoufakis - currently negotiating over the fate of his country's debt - is a student of "game theory". But what is it, asks Chris Stokel-Walker.

"Some commentators rushed to presume that as Greece's new finance minister I was busily devising bluffs, stratagems and outside options, struggling to improve upon a weak hand," wrote Varoufakis in the New York Times this week. "If anything, my game-theory background convinced me that it would be pure folly to think of the current deliberations between Greece and our partners as a bargaining game to be won or lost via bluffs and tactical subterfuge."

Game theory can be described as the mathematical study of decision-making, of conflict and strategy in social situations.

It helps explain how we interact in key decision-making processes.

Imagine you're buying a car from a dealership.

The dealer wants to sell the car, and has a fixed price beyond which he cannot drop without making a loss. You want to buy the car, but get the best price. The car dealer bases his negotiating tactics on the fact that you want the car, that he could sell the car to another customer, that he'll win praise for selling the car, and the break-even point beyond which he cannot negotiate down.

What is a game?

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- <u>A game</u> is an interaction between players, such as individuals or firms.
- In a game, each player makes decisions that affect the outcomes of others.
- <u>An action</u> is a specific move made at a particular stage of the game.
    - Example: a firm decides how much output to produce in the current period.
- <u>A strategy</u> is a plan that tells a player what actions to take.
- Players are usually assumed to be <u>rational</u> who try to maximise their payoff.
- They choose the strategy that gives them the highest benefit.

What is a game?

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- A payoff is the benefit a player receives from the outcome of the game.
    - For firms, payoff is usually profit.
    - For individuals, payoff may be income, utility, or another benefit.
- A payoff function shows how each player’s payoff depends on the strategies chosen by all players.
- This means one player’s result depends on both their own decision and the decisions of others.

Rules of the game

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- The rules of the game explain how the game is played.
- They include:
  - who moves first
  - when players move
  - what actions are possible
  - any other conditions of the game
- Information is very important in game theory. To analyse a game, we need to know how much information players have. The amount of information affects the strategies players choose.

Simultaneous-Move, Non-Repeated Interaction

- A one-shot game is played only once.
- Complete information. All players know:
  - the possible strategies
  - the possible payoffs
- A simultaneous game is one where players choose their actions at the same time. Players must make decisions without observing rivals first.
  - Example: Firms competing for a contract often face a simultaneous one-shot game. Each firm submits its bid independently. The firms do not observe each other’s bids before choosing. The contract is usually given to the lowest bidder. After that, the game ends.

Simultaneous-Move, Non-Repeated Interaction

- In this example, there are two firms:
    - identical costs
    - identical products
    - identical demand
- Each firm must choose between two prices: £2.00 and £1.80

Profits for firms X and Y at different prices

		X's price	
		£2.00	£1.80
Y's price	£2.00	A £10m, £10m	B £5m, £12m
	£1.80	C £12m, £5m	D £8m, £8m

A dominant strategy

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- A dominant strategy is the best strategy for a firm, no matter what its rival does.

Simultaneous-Move, Non-Repeated Interaction

- If both firms charge £2.00: each firm earns £10 million and total industry profit = £20 million. This is the outcome in cell A
- Firm X must think about what Firm Y might do.
- Firm Y has two possible actions: keep price at £2.00 or cut price to £1.80
- Firm X compares its profits under each case before deciding.

Simultaneous-Move, Non-Repeated Interaction

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If Firm Y keeps price at £2.00

- If Firm X also keeps price at £2.00: Firm X earns £10 million
- If Firm X cuts price to £1.80:Firm X earns £12 million
- Best response for Firm X: cut price to £1.80

If Firm Y cuts price to £1.80

- If Firm X keeps price at £2.00: Firm X earns £5 million
- If Firm X also cuts price to £1.80: Firm X earns £8 million
- Best response for Firm X:cut price to £1.80

Simultaneous-Move, Non-Repeated Interaction

- Firm X chooses £1.80 in both cases. So, cutting price to £1.80 is Firm X's dominant strategy. Firm Y will think in the same way. Therefore, both firms cut price to £1.80.
- Both firms end up charging £1.80. Each firm earns £8 million. This outcome is shown in cell D
- This outcome is called a Nash equilibrium.

Profits for firms X and Y at different prices

		X's price	
		£2.00	£1.80
Y's price	£2.00	A £10m, £10m	B £5m, £12m
	£1.80	C £12m, £5m	D £8m, £8m

Simultaneous-Move, Non-Repeated Interaction

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- A Nash equilibrium is a situation where: each firm chooses the best strategy given what the rival does, neither firm has an incentive to change its decision.
- Without collusion, both firms stay at this outcome
- In the Nash equilibrium, each firm earns £8 million. But if both firms charged £2.00:each would earn £10 million. So, firms are worse off in the Nash equilibrium than under collusion

Prisoners’ dilemma

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- Even if both firms would benefit from keeping the higher price, each firm has an incentive to cheat by cutting price.
- This leads both firms to a lower-profit outcome.
- This situation is known as the **prisoners’ dilemma**

Prisoners’ dilemma

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- The prisoners’ dilemma is a famous example of game theory outside economics.
- It shows how people acting in their own self-interest may end up with a worse outcome.
- The example involves Nigel and Amanda, who are arrested for serious fraud.
- They are questioned separately. Each has two choices: confess or stay silent. Their sentence depends on what both prisoners decide.

Prisoners' dilemma

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- If both stay silent: each gets 1 year in prison. If one confesses and the other stays silent: the person who confesses gets 3 months and the other could get up to 10 years. If both confess each gets 3 years in prison.
- If Amanda stays silent, Nigel does better by confessing. If Amanda confesses, Nigel still does better by confessing. So, Nigel's best response is always to confess
- Amanda reasons in exactly the same way. Her best response is also to confess. Therefore, both prisoners confess

The prisoners' dilemma

		Amanda's alternatives	
		Not confess	Confess
Nigel's alternatives	Not confess	A Each gets 1 year	B Nigel gets 10 years Amanda gets 3 months
	Confess	C Nigel gets 3 months Amanda gets 10 years	D Each gets 3 years

Prisoners’ dilemma

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- When both confess, they reach a Nash equilibrium
- Each is making the best decision based on what they think the other will do. Neither has an incentive to change their decision alone
- But this outcome is worse for both than if both stayed silent

Why collusion is difficult

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- The best joint outcome is for both prisoners to stay silent
- But each has a strong incentive to cheat and confess
- The police try to prevent cooperation by:
  - keeping them in separate cells
  - making each believe the other will confess

Real-world example – too much advertising

- Firms may increase advertising to get ahead of rivals
- They may also advertise more to avoid falling behind
- For each firm, spending more on advertising may seem like the best choice
- But the final result may be:
  - too much advertising
  - higher costs for all firms
  - little or no extra sales overall

DOMINANT STRATEGIES

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- Unfortunately, not every game has a dominant strategy for each player. To see this, let's change our advertising example slightly.

		Firm B	
		Advertise	Don't advertise
Firm A	Advertise	10, 5	15, 0
	Don't advertise	6, 8	20, 2

A more complicated game

		X's price	
		£25	£19
Y's price	£20	A £6m, £6m	B £2m, £5m
	£15	C £4m, £3m	D £4m, £4m

Repeated simultaneous games infinitely

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- Many firms do not compete only once. They compete again and again over time.
- This means today's decision can affect future behaviour and profits
    - Examples: supermarkets changing prices regularly, fast food firms adjusting prices, Apple and Samsung launching new phones each year
- Firms can observe what rivals did in earlier periods. They can use this information when choosing their next action.
- This creates a link between the short run and the long run
- Firms may avoid actions today if they fear punishment later

Repeated simultaneous games

- In a one-shot prisoners' dilemma, both players usually choose the Nash equilibrium. This gives a worse outcome for both players.
- But in a repeated game, cooperation may become possible. Players may cooperate if they care about future outcomes.

The grim trigger strategy

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- Grim trigger means: cooperate at first but if the rival cheats once, punish forever. After one act of cheating, the player switches to a permanent punishment strategy. This creates a strong threat against cheating.
- A player may gain in the short run by cheating. But they may lose much more in the long run because punishment continues forever. So players may decide that cooperation is the better choice
- Grim trigger helps support cooperation. It works because players want to avoid permanent retaliation. It is effective when players expect the game to continue for a long time

The tit-for-tat strategy

- Tit-for-tat means: start by cooperating then copy whatever the rival did in the previous round. If the rival cooperates, you cooperate. If the rival cheats, you punish by cheating next round.

- Tit-for-tat includes punishment. But it also includes forgiveness. If the rival returns to cooperation, you cooperate again. This can make it easier to restore cooperation

Application to pricing

- In pricing, firms may use tit-for-tat behaviour. A firm may decide not to cut price first but to cut price if a rival cuts price. This is a way of discouraging price cuts.
- Repeated games can help explain <u>tacit collusion</u>. Firms may avoid aggressive actions because they fear retaliation. This can help firms keep prices higher without a formal agreement

Finitely repeated games

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- A finitely repeated game is a game played a fixed number of times
  - Example: the game is played for 10 rounds only. Players know exactly when the game will end
- In the last round, there is no future interaction. So players cannot be punished or rewarded later. This removes the incentive to cooperate. The game becomes like a one-shot game.
- In the final round, both players choose the strategy that is best for themselves at that moment. In the prisoners' dilemma, this means both players choose confess. So the Nash equilibrium returns in the last round

Sequential games

- A sequential game is a game where firms make decisions one after another.
- The order of play matters. Firms can observe earlier decisions before making their own choices
- In real markets, firms often do not act at the same time. One firm may move first and rivals respond later.
    - Example: a firm announces new prices, then competitors react

Sequential games

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- Boeing and Airbus must choose which aircraft to produce: a 400-seater or a 500-seater. The market is not large enough for both firms to profitably produce the same type.
- If only one firm produces the 400-seater: profit = £50 million. If only one firm produces the 500-seater:profit = £30 million. If both firms produce the same version: each makes a £10 million loss.
- A decision tree (or game tree) shows: the sequence of decisions, the choices available to each firm and the final outcomes. It is used to analyse sequential games.

Sequential games

If Boeing chooses the 500-seater first, Airbus then decides how to respond

- If Airbus also chooses the 500-seater: both firms make a loss
- If Airbus chooses the 400-seater: Boeing earns £30 million- Airbus earns £50 million.

So Airbus's best response is to choose the 400-seater.

If Boeing chooses the 400-seater first, Airbus again decides how to respond

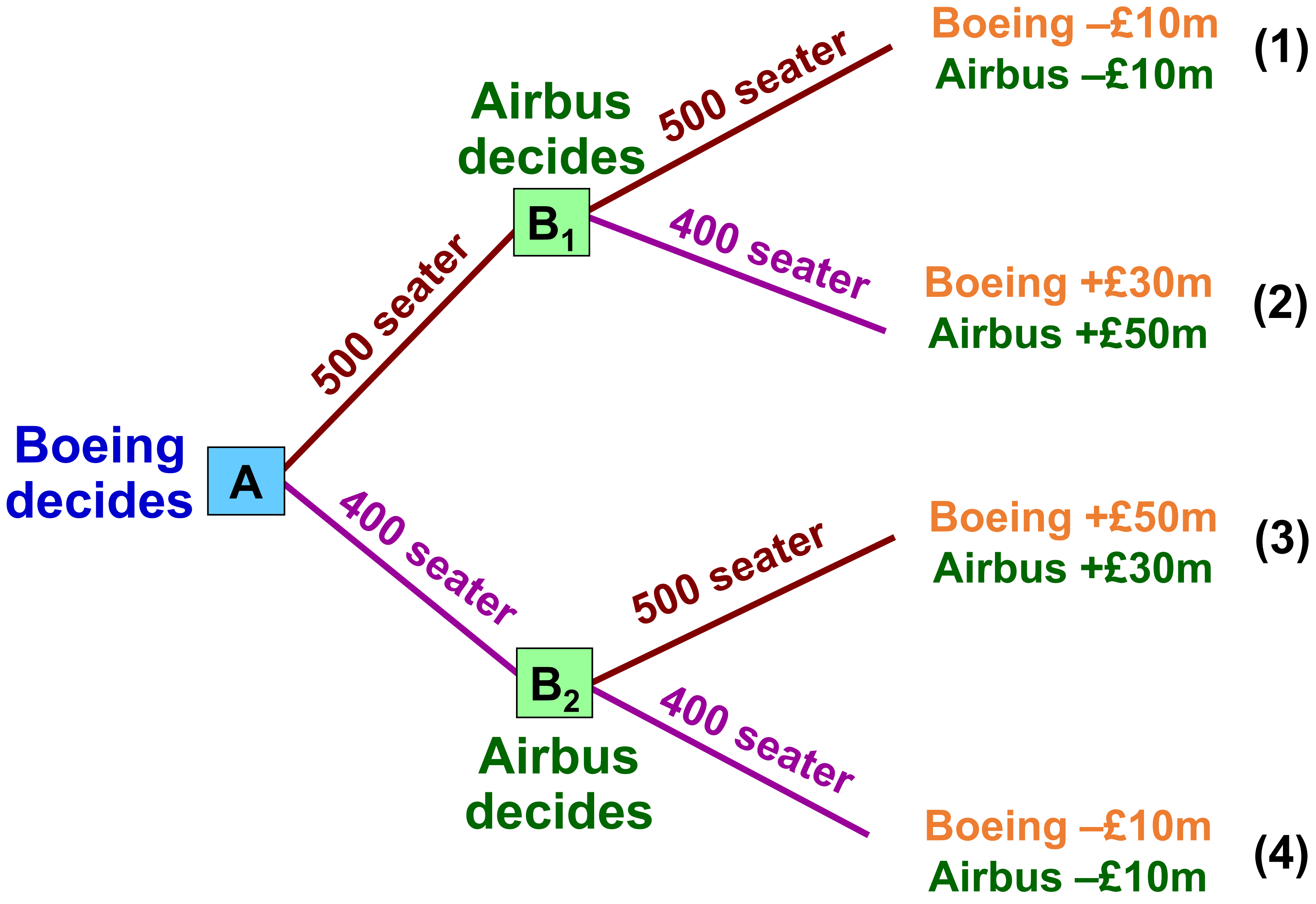
- If Airbus also chooses the 400-seater: both firms make a loss
- If Airbus chooses the 500-seater: Boeing earns £50 million and Airbus earns £30 million

So Airbus's best response is to choose the 500-seater

Sequential games

- Boeing thinks ahead and predicts Airbus's response.
- Boeing chooses the 400-seater-Airbus then chooses the 500-seater.
- This is the Nash equilibrium
- Final outcome: Boeing earns £50 million and Airbus earns £30 million

A decision tree



First-mover advantage

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- The firm that moves first may gain a first-mover advantage
- By moving first, Boeing secures the more profitable market
- Airbus is forced to accept the less profitable option
- First-mover advantage depends on factors such as:
    - research and development progress
    - production capacity
    - speed of innovation
- The firm that acts first may shape the market outcome

Credible threats and promises

- In sequential games, firms may use threats or promises to influence rivals' decisions. The main issue is whether the threat or promise is credible.
- A credible threat is one that the firm is actually willing to carry out. A threat is credible if it is in the firm's own interest to carry it out
- A threat is non-credible if the firm would not really act that way when the time comes. Firms will usually ignore threats that are not believable

Credible threats and promises

- In sequential games, threats are often made by the second mover. The second mover tries to influence what the first mover does. The first mover must decide whether to believe the threat.

- The first mover looks ahead in the game tree. It asks: what will the second mover do later? is the threat believable? The first mover's choice depends on the answer

Questions